

# Proof that $\sqrt{11}$ is Irrational

## Problem

Prove that  $\sqrt{11}$  is irrational.

## Proof

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*Proof.* We first record two elementary facts needed in the proof.

**Fact 1 (Lowest-terms representation).** If  $x \in \mathbb{Q}$ , then there exist integers  $a, b$  with  $b \neq 0$  and  $\gcd(a, b) = 1$  such that  $x = \frac{a}{b}$ .

*Proof of Fact 1.* Since  $x \in \mathbb{Q}$ , there exist integers  $m, n$  with  $n \neq 0$  and  $x = \frac{m}{n}$ . Let  $d = \gcd(m, n) > 0$  and write  $m = da, n = db$  for integers  $a, b$ . Since  $n \neq 0$  we have  $b \neq 0$ , and  $x = \frac{a}{b}$ . If  $e$  is any positive common divisor of  $a$  and  $b$ , then  $de$  divides both  $m$  and  $n$ , so  $de \leq d$ , hence  $e \leq 1$ . Therefore  $\gcd(a, b) = 1$ .  $\square$

**Fact 2.** For any integer  $n$ , if  $11 \mid n^2$  then  $11 \mid n$ .

*Proof of Fact 2.* By the division algorithm,  $n \equiv r \pmod{11}$  for some  $r \in \{0, 1, 2, \dots, 10\}$ . Since  $11 \mid (n - r)$  we have  $n^2 - r^2 = (n - r)(n + r) \equiv 0 \pmod{11}$ , so  $n^2 \equiv r^2 \pmod{11}$ . Computing the squares of all residues modulo 11:

| $r$             | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|-----------------|---|---|---|---|---|---|---|---|---|---|----|
| $r^2 \pmod{11}$ | 0 | 1 | 4 | 9 | 5 | 3 | 3 | 5 | 9 | 4 | 1  |

The only residue satisfying  $r^2 \equiv 0 \pmod{11}$  is  $r = 0$ . Hence if  $11 \mid n^2$  then  $r = 0$ , so  $n \equiv 0 \pmod{11}$ , i.e.  $11 \mid n$ .  $\square$

Now suppose, for contradiction, that  $\sqrt{11}$  is rational. By Fact 1, write  $\sqrt{11} = \frac{a}{b}$  with  $a, b \in \mathbb{Z}$ ,  $b \neq 0$ , and  $\gcd(a, b) = 1$ . Squaring both sides and multiplying by  $b^2 \neq 0$  gives

$$a^2 = 11b^2.$$

Hence  $11 \mid a^2$ , and by Fact 2,  $11 \mid a$ . Write  $a = 11c$  for some integer  $c$ . Substituting:

$$(11c)^2 = 11b^2 \implies 121c^2 = 11b^2 \implies 11c^2 = b^2.$$

Hence  $11 \mid b^2$ , and by Fact 2 again,  $11 \mid b$ .

So 11 divides both  $a$  and  $b$ , contradicting  $\gcd(a, b) = 1$ . Therefore  $\sqrt{11}$  is irrational.  $\square$